Properties

① $\lim_{x\to x_0} f(x) = a$ $\lim_{x\to x_0} f(x) = b$ $\Leftrightarrow$ $a = b$

② If $\lim_{x\to x_0} f(x)$ exists $\Rightarrow$ $f(x)$ is bounded

③ If $f(x) \geq 0$, or $f(x) > 0$ $\Rightarrow$ $\lim f(x) \geq 0$

④ $\lim (f(x) \pm g(x)) = \lim f(x) \pm \lim g(x)$

$\lim c f(x) = c \cdot \lim f(x)$ $c \in \mathbb{R}$

$\lim (f(x) \cdot g(x)) = \lim f(x) \cdot \lim g(x)$

$\lim \frac{f(x)}{g(x)} = \frac{\lim f(x)}{\lim g(x)}$ given $g(x) \neq 0$, $\lim g(x) \neq 0$.

⑤ $f(x) \leq h(x) \leq g(x)$ $\forall x \in (x_0 - \varepsilon, x_0) \cup (x_0, x_0 + \varepsilon)$

$\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x)$ $\Rightarrow$ $\lim_{x\to x_0} h(x) = A$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$ $\lim_{x\to\infty} (1 + \frac{1}{x})^x = e$

1. Infinitesimal quantity 无穷小量

Defn: If f is well-defined on $(x_0 - \varepsilon, x_0) \cup (x_0, x_0 + \varepsilon)$, and

$$\lim_{x\to x_0} f(x) = 0$$

then f is an infinitesimal quantity as $x \to x_0$

E.g. 1 $\lim_{x\to 0} x^2 = 0$ $\lim_{x\to 0} \sin x = 0$ $\lim_{x\to\infty} \frac{1}{x} = 0$

Properties: ① if $\lim_{x\to x_0} f(x) = 0$, $\lim_{x\to x_0} g(x) = 0$ $\Rightarrow$ $\lim_{x\to x_0} (f(x) \pm g(x)) = 0$

$\Rightarrow$ $\lim_{x\to x_0} f(x) \cdot g(x) = 0$

② if $\lim_{x\to x_0} f(x) = 0$ $\lim_{x\to x_0} g(x) = A < M$ $\Rightarrow$ $\lim_{x\to x_0} f(x) \cdot g(x) = 0$

E.g: $\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0$

If $\lim_{x\to x_0} \frac{f(x)}{g(x)} = 0$, $\lim_{x\to x_0} f(x) = 0$ and $\lim_{x\to x_0} g(x) = 0$, then we say

f is higher order infinitesimal quantity than g. or

$$f(x) = o(g(x)) \quad (x \to x_0)$$

E.g. $\lim_{x\to 0} \frac{x^2}{x} = 0$ $\qquad \lim_{x\to 0} \frac{1-\cos x}{\sin x} = \lim_{x\to 0} \tan\frac{x}{2} = 0$

If $\lim_{x\to x_0} f(x) = 0$ and $\lim_{x\to x_0} g(x) = 0$, but $\lim_{x\to x_0} \frac{f(x)}{g(x)} = C \neq 0$, we say

f and g have the same order of infinitesimal quantity. or

$$f(x) = O(g(x)) \quad (x \to x_0)$$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$ $\qquad \lim_{x\to 0} \frac{x^2}{1-\cos x} = 2 \Rightarrow$ $\quad 1-\cos x = O(x^2) \ (x\to 0)$

In Particular, if $\lim_{x\to x_0} \frac{f(x)}{g(x)} = 1$, we say f and g are equivalent infinitesimal quantity, or

$$f(x) \sim g(x) \quad (x \to x_0)$$

Thm: Suppose that f, g, h are well-defined on $(x_0-\varepsilon)\cup(x_0+\varepsilon)$ and $f \sim g \ (x \to x_0)$

① $\lim_{x\to x_0} f(x)\cdot h(x) = A \Rightarrow \lim_{x\to x_0} g(x)\cdot h(x) = A$

② $\lim_{x\to x_0} \frac{h(x)}{f(x)} = B \Rightarrow \lim_{x\to x_0} \frac{h(x)}{g(x)} = B$

E.g. $\lim_{x\to 0} \frac{\arctan x}{\sin 4x}$

Commonly used equivalent infinitesimal quantity.

when $x \to 0$. $\sin x \sim x$ $\quad \arcsin x \sim x$

$\tan x \sim x$ $\quad \arctan x \sim x$

$\ln(1+x) \sim x$ $\quad e^x - 1 \sim x$

$a^x - 1 \sim x \ln a$

$1 - \cos x \sim \frac{1}{2}x^2$

$\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$

when $x \to 1$ $\quad \ln x \sim x - 1$

E.g. $\lim_{x \to 0} \frac{\arctan x}{\sin 4x} = \lim_{x \to 0} \frac{x}{4x} = \frac{1}{4}$ $\quad \arctan x \sim x \quad \sin x \sim x$

$\sin 4x \sim 4x$

E.g. $\lim_{x \to 0} \frac{\tan x - \sin x}{\sin(x^3)} = \lim_{x \to 0} \frac{\frac{\sin x}{\cos x}(1 - \cos x)}{\sin(x^3)} = \lim_{x \to 0} \frac{1}{\cos x} \frac{x \cdot \frac{1}{2}x^2}{x^3} = \frac{1}{2}$

$\sin(x^3) \sim x^3 \quad \sin x \sim x \quad (1 - \cos x) \sim \frac{1}{2}x^2$

Infinite quantity 无穷大量

Defn. If f is well-defined $(x_0 - \varepsilon, x_0) \cup (x_0, x_0 + \varepsilon)$, and

$$\lim_{x \to x_0} |f(x)| = +\infty$$

then we say f is an infinite quantity as $x \to x_0$.

E.g. $\lim_{x \to 0} \frac{1}{x^2} = +\infty \quad \lim_{x \to 0^-} \frac{1}{x} = -\infty \quad \lim_{x \to 0^+} \frac{1}{x} = +\infty$

Properties $\lim_{x \to x_0} f(x) = +\infty \quad \lim_{x \to x_0} g(x) = +\infty \quad \lim_{x \to x_0} h(x) = A$

$\Rightarrow \lim_{x \to x_0} (f(x) + g(x)) = +\infty \quad \lim_{x \to x_0} f(x) \cdot g(x) = +\infty$

$\lim_{x \to x_0} (f(x) + h(x)) = +\infty \quad \lim_{x \to x_0} (f(x) \cdot h(x)) = +\infty$

$$\lim_{x\to\frac{\pi}{2}} 2(\sin x - \cos x - x^2) = 2 - \frac{\pi^2}{2}$$

$$\lim_{x\to 0} \frac{x^2-1}{2x^2-x-1} = 1$$

$$\lim_{x\to 1} \frac{x^2-1}{2x^2-x-1} = \lim_{x\to 1} \frac{(x+1)(x-1)}{(x-1)(2x+1)} = \lim_{x\to 1} \frac{x+1}{2x+1} = \frac{2}{3}$$

$$\lim_{x\to 0} \frac{(x-1)^3 + (1-3x)}{x^2+2x^3} = \frac{x^3-3x^2+3x-1+1-3x}{} = \lim_{x\to 0} \frac{x^2(x-3)}{x^2(1+2x)} = -3$$

$$\lim_{x\to 1} \frac{x^n-1}{x^m-1} = \lim_{x\to 1} \frac{\cancel{(x-1)}(x^{n-1}+x^{n-2}+\cdots+x+1)}{\cancel{(x-1)}(x^{m-1}+x^{m-2}+\cdots+x+1)} = \frac{n}{m}$$

$$\lim_{x\to 4} \frac{\sqrt{1+2x}-3}{\sqrt{x}-2} = \lim_{x\to 4} \frac{(\sqrt{1+2x}-3)(\sqrt{1+2x}+3)}{(\sqrt{x}-2)(\sqrt{1+2x}+3)} = \lim_{x\to 4} \frac{2\cancel{(x-4)}(\sqrt{x}+2)}{\cancel{(x-4)}(\sqrt{1+2x}+3)} = \frac{4}{3}$$

$$\lim_{x\to 0} \frac{\sqrt{a^2+x}-a}{x} = \lim_{x\to 0} \frac{a^2+x-a^2}{x(\sqrt{a^2+x}+a} = \frac{1}{2a}$$

$(a>0)$

2. Find the limits by using squeeze theorem

$$\lim_{x\to -\infty} \frac{x-\cos x}{x} \qquad 1-\frac{1}{x} < 1-\frac{\cos x}{x} < 1+\frac{1}{x}$$

$$\lim_{x\to +\infty} \frac{x\sin x}{x^2-4} = 0 \qquad \frac{-x}{x^2-4} < \frac{x\sin x}{x^2-4} < \frac{x}{x^2-4}$$

$$\lim_{x\to\infty} \frac{[x]}{x} \qquad \frac{x-1}{x} \le \frac{[x]}{x} \le \frac{x+1}{x}$$

Since $\lim_{x\to\infty} \frac{x-1}{x} = \lim_{x\to\infty} \frac{x+1}{x} = 1$, we have $\lim_{x\to\infty} \frac{[x]}{x} = 1$

3. Based on $\lim_{x\to 0} \frac{\sin x}{x} = 1$, find the following limits.

$$\lim_{x\to 0} 2\frac{\sin 2x}{2x} = 2$$

$$\lim_{x\to 0} \frac{\sin x^3}{\sin^2 x} \quad \cancel{\frac{x^2}{\sin^2 x}}^{\to 1} \cdot \cancel{\frac{\sin x^3}{x^3}}^{1} \cdot \cancel{x}^{\to 0} = 0$$

$$\lim_{x\to \frac{\pi}{2}} \frac{\cos x}{x-\frac{\pi}{2}} \quad x-\frac{\pi}{2} = y$$

$$= \frac{\sin y}{y} = 1$$

$$\lim_{x\to 0} \frac{\tan x}{x} = \frac{1}{\cos x} = 1$$

$$\lim_{x\to 0} \frac{\tan x - \sin x}{x^3} = \lim_{x\to 0} \frac{\sin x - \sin x \cdot \cos x}{x^3 \cos x} = \lim_{x\to 0} \frac{\sin x}{x} \frac{(1-\cos x)}{x^2}$$

$$\lim_{x\to 0} \frac{\arctan x}{x} \quad y = \arctan x \quad x = \tan y \quad \lim_{y\to 0} \frac{y}{\tan y}$$

$$\lim_{x\to +\infty} x \sin\frac{1}{x} \quad y = \frac{1}{x}$$

$$\lim_{x\to a} \frac{\sin^2 x - \sin^2 a}{x-a}$$

$$\sin a + \sin b = 2\sin\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right)$$

$$\sin a - \sin b = 2\cos\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right)$$

$$\sin 2a = 2\sin a \cdot \cos a$$

$$= \lim_{x\to a} \frac{(\sin x + \sin a)(\sin x - \sin a)}{x-a}$$

$$= \lim_{x\to a} \frac{2\sin\left(\frac{x+a}{2}\right)\cos\left(\frac{x-a}{2}\right) \cdot 2\sin\left(\frac{x-a}{2}\right)\cos\left(\frac{x+a}{2}\right)}{x-a}$$

$$= \lim_{x\to a} \frac{\sin(x-a)\sin(x+a)}{x-a} = \sin 2a$$

4. Given $\lim_{x\to\infty}(1+\frac{1}{x})^x = e$, find the limits of the following.

$$\lim_{x\to 0}(1+2x)^{\frac{1}{x}} = \lim_{x\to 0}(1+2x)^{2\cdot\frac{1}{2x}} = \lim_{x\to 0}\left((1+2x)^{\frac{1}{2x}}\cdot(1+2x)^{\frac{1}{2x}}\right) = e\cdot e$$

$$\lim_{x\to 0}(1+\alpha x)^{\frac{1}{x}} \quad y=\frac{1}{\alpha x} \quad x=\frac{1}{\alpha y} \quad \lim_{|y|\to\infty}\left(1+\frac{1}{y}\right)^{\alpha y} = e^{\alpha}$$

$$\lim_{x\to 0}(1+\tan x)^{\cot x} \quad y=\cot x \quad \tan x=\frac{1}{y} \quad \lim_{y\to+\infty}\left(1+\frac{1}{y}\right)^{y} = e$$

$$\lim_{x\to 0}\left(\frac{1+x}{1-x}\right)^{\frac{1}{x}} = \lim_{x\to 0}\left(\frac{1-x+2x}{1-x}\right)^{\frac{2}{x}} = \lim_{x\to 0}\left(1+\frac{2x}{1-x}\right)^{\frac{1}{x}} \quad y=\frac{1-x}{2x} \quad x=\frac{1}{2y+1}$$

$$\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{2y+1} = \lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{2y}\cdot\left(1+\frac{1}{y}\right) = e^2\cdot 1$$

$$\lim_{x\to+\infty}\left(\frac{3x+2}{3x-1}\right)^{2x-1} = \lim_{x\to+\infty}\left(1+\frac{1}{x-\frac{1}{3}}\right)^{2x-1} \quad y=x-\frac{1}{3} \quad x=y+\frac{1}{3}$$

$$\lim_{y\to+\infty}\left(1+\frac{1}{y}\right)^{2y-\frac{1}{3}} = e^2\cdot 1$$

$$\lim_{x\to+\infty}\left(1+\frac{\alpha}{x}\right)^{\beta x} \quad y=\frac{1}{\alpha}x \quad x=\alpha y \quad \lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{\beta\cdot\alpha y} = e^{\alpha\beta}$$

5 Use the equivalent infinitesimal quantity to find the following limits.

$$\lim \frac{x}{x+A} = 1$$

$$\lim_{x\to\infty}\frac{x\arctan\frac{1}{x}}{x-\cos x} = \lim_{x\to\infty}\frac{x}{x-\cos x}\cdot\lim_{x\to\infty}\arctan\frac{1}{x} = 1\cdot 0$$

$$\lim_{x\to 0}\frac{\sqrt{1+x^2}-1}{1-\cos x} = \lim_{x\to 0}\frac{\frac{1}{2}x^2}{1-\cos x}\cdot\lim_{x\to 0}\frac{\sqrt{1+x^2}-1}{\frac{1}{2}x^2} = \lim_{x\to 0}\frac{1+x^2-1}{\frac{1}{2}x^2\left(\sqrt{1+x^2}+1\right)} = 1$$